Lie's Third Theorem

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Preliminaries

Definition 1.1

Let G be a matrix lie group.The lie algebra of G denoted as $\mathfrak{g}$ is the set of all matrices X s.t $e^{tX} \in G$ for all real numbers t .

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Theorem 1.1

The function

$$\log A = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{(A-I)^m}{m}$$

is defined and continuous on the set of all $n \times n$ complex matrices A with $\|A - I\| < 1$. For all A with $\|A - I\| < 1$,

$$e^{\log A} = A$$

For all X with $\|X\| < \log 2$, $\left\|e^X - I\right\| < 1$ and

$$\log e^X = X$$

Lemma 1

Assume $G \subset \mathrm{GL}(m;\mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Assume B_n is a sequence in G and $B_n \to I$. Let $Y_n = \log B_n$, which is defined for all large n. Suppose that Y_m is nonzero for all n and $Y_n / \|Y_n\| \to Y \in M_n(\mathbb{C})$. Then $Y \in \mathfrak{g}$.

Lemma 1

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Proof

For any $t \in \mathbb{R}$, we have $(t/\|Y_n\|)\, Y_n \to tY$. Observe that since $B_n \to I$, we get $\|Y_n\| \to 0$. Thus, we can find integers k_n such that $k_m \|Y_m\| \to t$. Denote $\|Y_n\|$ as a_n. Then, take $k_n = \lfloor \frac{t}{a_n} \rfloor$.
Using squeeze theorem for the inequality ,

$$a_n\left(\frac{t}{a_n} - 1\right) \leq a_n \left\lfloor \frac{t}{a_n} \right\rfloor \leq a_n\left(\frac{t}{a_n}\right)$$

we get that $k_n a_n \to t$.

Proof Continued

We get,

$$e^{k_n Y_n} = \exp\left[(k_n \|Y_n\|)\frac{Y_n}{\|Y_n\|}\right] \to e^{tY}$$

But,

$$e^{k_n Y_n} = \left(e^{Y_n}\right)^{k_n} = (B_n)^{k_n} \in G$$

and G is closed. So, $e^{tY} \in G$. Hence, $Y \in \mathfrak{g}$.

Theorem 1.2

For $\varepsilon \in (0, \log 2)$, let $U_\varepsilon = \{X \in M_n(\mathbb{C}) \mid \|X\| < \varepsilon\}$ and let $V_\varepsilon = \exp(U_\varepsilon)$. Suppose $G \subset \mathrm{GL}(n;\mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Then there exists $\varepsilon \in (0, \log 2)$ such that for all $A \in V_\varepsilon$, A is in G if and only if $\log A$ is in $\mathfrak{g}$.

Theorem 1.2

For $\varepsilon \in (0, \log 2)$, let $U_\varepsilon = \{X \in M_n(\mathbb{C}) \mid \|X\| < \varepsilon\}$ and let $V_\varepsilon = \exp(U_\varepsilon)$. Suppose $G \subset \mathrm{GL}(n; \mathbb{C})$ is a matrix Lie group with Lie algebra $\mathfrak{g}$. Then there exists $\varepsilon \in (0, \log 2)$ such that for all $A \in V_\varepsilon, A$ is in G if and only if $\log A$ is in $\mathfrak{g}$.

Proof

Observe that for all $A \in V_\varepsilon$, $A = e^X$ given $||X|| < \log 2$, which implies $||A - I|| < 1$. So, $\log(e^X) = X$. Now if $X = \log A$ is in $\mathfrak{g}$, then $A = e^{\log A} = e^X$ is in G. So, need to prove the other direction.
Let us veiw $M_n(\mathbb{C})$ as $\mathbb{C}^{n^2} \cong \mathbb{R}^{2n^2}$ and say D be the orthogonal complement of $\mathfrak{g}$ with respect to the normal inner product on $\mathbb{R}^{2n^2}$.
Consider the map $\Phi : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ given by

$$\Phi(Z) = e^X e^Y$$

where $Z = X + Y$ with $X \in \mathfrak{g}$ and $Y \in D$. We claim that

Proof Continued

derivative of ϕ at origin $D_0(\Phi) = I$. Take $h = h_1 + h_2$ where $h_1 \in X, h_2 \in Y$. Then,

$$\begin{aligned}
\lim_{h\to 0} \frac{\left\| e^{h_1} \cdot e^{h_2} - I - Ih \right\|}{\|h\|} &= \lim_{h\to 0} \| h_1 h_2 + \frac{h_1^2}{2!} + \frac{h_2^2}{2!} + \cdots \cdot \| \frac{1}{\|h\|} \\
&\leq \lim_{h\to 0} \left(\|h\|^2 + \frac{\|h\|^2}{2!} + \frac{\|h\|^2}{2!} + \cdots \right) \frac{1}{\|h\|} \\
&= \lim_{h\to 0} \left(\|h\| + \frac{\|h\|}{2!} + \frac{\|h\|}{2!} + \cdots \right) = 0
\end{aligned}$$

Since the derivative of Φ at the origin is invertible, applying inverse function theorem we get Φ has a continuous local inverse, defined in a neighborhood of I.

We need to prove that for some ε, if $A \in V_\varepsilon \cap G$, then $\log A \in \mathfrak{g}$. If this is not true, we can find a sequence A_m in G such that $A_m \to I$ as $m \to \infty$ and such that for all $m, \log A_m \notin \mathfrak{g}$.

Proof Continued

Using the local inverse of the map Φ, we can write A_m (for all sufficiently large m) as

$$A_m = e^{X_m} e^{Y_m}, \quad X_m \in \mathfrak{g}, Y_m \in D$$

with X_m ,$Y_m \to 0$ as $m \to \infty$. We must have $Y_m \neq 0$, otherwise we would get $\log A_m = X_m \in \mathfrak{g}$. Since e^{X_m}, $A_m \in G$, we see that

$$B_m := e^{-X_m} A_m = e^{Y_m} \in G.$$

Since the unit sphere in D is compact, we can choose a subsequence of the Y_m so that $Y_m / \|Y_m\|$ converges to some $Y \in D$, with $\|Y\| = 1$. By the lemma (1.2), $Y \in \mathfrak{g}$. This is a contradiction, as D is the orthogonal complement of $\mathfrak{g}$. So, there must be some ε such that $\log A \in \mathfrak{g}$ for all A in $V_\varepsilon \cap G$.

Derivative of exponential

Theorem 1.3

Let X(t) be a smooth matrix valued function. Then

$$\frac{d}{dt}e^{X(t)} = e^{X(t)}\left\{\frac{I - e^{-ad_{X(t)}}}{ad_{X(t)}}\left(\frac{dX}{dt}\right)\right\}$$

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Proof

Write X instead of $X(t)$, and let $Y = X'$ be the derivative. Then ,

$$\begin{aligned}
I' = (X^0)' &= 0 \ , \\
(X^1)' &= Y \ , \\
(X^2)' &= YX + XY \ , \\
(X^3)' &= YXX + XYX + XXY \ , \\
(\ X^{n+1}\ )' &= YX^n + XYX^{n-1} + \cdots + X^nY = \sum_{\substack{j,k\geq 0 \\ j+k=n}} X^jYX^k \ .
\end{aligned}$$

Proof Continued

$$
\begin{aligned}
\frac{d}{dt}e^{X(t)} &= Y + \frac{1}{2!}(YX + XY) + \frac{1}{3!}(YX^2 + XYX + X^2Y) + \cdots \\
&= \sum_{k=0}^{\infty}\sum_{r=0}^{k} \frac{1}{(k+1)!} X^r Y X^{k-r} \\
&= \sum_{k=0}^{\infty}\sum_{r=0}^{k} \frac{B(k-r+1, r+1)}{r!(k-r)!} X^r Y X^{k-r} \\
&= \sum_{k=0}^{\infty}\sum_{r=0}^{k} \frac{\int_0^1 s^r (1-s)^{k-r} ds}{r!(k-r)!} X^{k-r} Y X^r \\
&= \int_0^1 \sum_{k=0}^{\infty}\sum_{r=0}^{k} \frac{(1-s)^{k-r} s^r}{r!(k-r)!} X^{k-r} Y X^r ds \quad (D.C.T) \\
&= \int_0^1 \left(\sum_{n=0}^{\infty} \frac{1}{r!}(1-s)^r X^r \right) Y \left(\sum_{n=0}^{\infty} \frac{1}{r!} s^r X^r \right) ds
\end{aligned}
$$

Proof Continued

$$= \int_0^1 e^{(1-s)X} Y e^{sX} ds.$$

Thus,

$$\begin{aligned}
\int_0^1 e^{(1-s)X} Y e^{sX} ds &= e^X \int_0^1 Ad_{e^{-sX}}(Y) ds \\
&= e^X \int_0^1 e^{-ad_{sX}}(Y) ds \\
&= e^X \left(\int_1^1 e^{-ad_{sX}} ds \right) (Y) \\
&= e^X \left(\int_0^1 \sum_{k=0}^{x} \frac{(-1)^k s^k}{k!} (ad_X)^k ds \right) (Y) \\
&= e^X \left\{ \frac{1 - e^{-ad_X}}{ad_X} (Y) \right\}
\end{aligned}$$

The goal of this section is to compute $\log(e^X e^Y)$ in terms of their commutators when X, Y are small. Consider the function

$$g(z) = \frac{\log z}{1 - \frac{1}{z}}$$

which is defined and holomorphic in the disk $\{|z-1| < 1\}$. Thus, $g(z)$ can be expressed as a series

$$g(z) = \sum_{m=0}^{\infty} a_m (z-1)^m = g(z) = 1 + \frac{1}{2}(z-1) - \frac{1}{6}(z-1)^2 + \frac{1}{12}(z-1)^3 - \cdots .$$

with radius of convergence one. If V is a finite-dimensional vector space over $\mathbb{C}$ then for any operator A on V with $\|A - I\| < 1$, we can define

$$g(A) = \sum_{m=0}^{\infty} a_m (A - I)^m$$

The Baker-Campbell-Haursdorff Formula

Theorem 2.1

For all $n \times n$ complex matrices X and Y with $\|X\|$ and $\|Y\|$ sufficiently small, we have

$$\log\left(e^X e^Y\right) = X + \int_0^1 g\left(e^{ad_X} e^{tad_Y}\right)(Y)dt$$

NOTE : Observe that $e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}$ and hence, also $g\left(e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}\right)$ are linear operators on the space $M_n(\mathbb{C})$ of all $n \times n$ complex matrices which is applied on the matrix Y. Here X and Y are assumed small implies that $e^{\mathrm{ad}X} e^{t\mathrm{ad}_Y}$ is close to the identity operator on $M_n(\mathbb{C})$ for $0 \leq t \leq 1$, so that $g\left(e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}\right)$ is well defined.

Proof

For sufficiently small X and Y in $M_n(\mathbb{C})$, let

$$Z(t) = \log\left(e^X e^{tY}\right)$$

for $0 \leq t \leq 1$. We will compute $Z(1)$. Since $e^{Z(t)} = e^X e^{tY}$, we have

$$e^{-Z(t)} \frac{d}{dt} e^{Z(t)} = \left(e^X e^{tY}\right)^{-1} e^X e^{tY} Y = Y$$

On the other hand, by Theorem (1.4),

$$e^{-Z(t)} \frac{d}{dt} e^{Z(t)} = \left\{ \frac{I - e^{-\mathrm{ad}_{Z(t)}}}{\mathrm{ad}_{Z(t)}} \right\} \left(\frac{dZ}{dt} \right)$$

Hence,

$$\left\{ \frac{I - e^{-\,\mathrm{ad}_{Z(t)}}}{\mathrm{ad}_{Z(t)}} \right\} \left(\frac{dZ}{dt} \right) = Y$$

Proof Continued

Now, if X and Y are small enough, $Z(t)$ will also be small, so that $T = \left[I - \ e^{-\mathrm{ad}_{Z(t)}}\right] / \mathrm{ad}_{Z_{(t)}}$ will be close to the identity and so its eigenvalues close to 1 as $|\lambda - 1| < \|T - I\|$ and hence invertible. So, we get

$$\frac{dZ}{dt} = \left\{ \frac{I - e^{-\mathrm{ad}_{Z(t)}}}{\mathrm{ad}_{Z(t)}} \right\}^{-1} (Y)$$

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Proof Continued

Also applying the Ad homomorphism formula to the equation $e^{Z(t)} = e^X e^{tY}$, we get :

$$\begin{aligned} \mathrm{Ad}_{e^{Z(t)}} &= \mathrm{Ad}_{e^X}\,\mathrm{Ad}_{e^{tY}} \\ \implies e^{\mathrm{ad}_{Z(t)}} &= e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y} \\ \implies \mathrm{ad}_{\mathrm{Z}(t)} &= \log\left(e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}\right). \end{aligned}$$

Putting the last relations give us:

$$\frac{dZ}{dt} = \left\{ \frac{I - \left(e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}\right)^{-1}}{\log\left(e^{\mathrm{ad}_X} e^{t\mathrm{ad}_Y}\right)} \right\}^{-1} (Y)$$

Now, notice that

$$g(z) = \left\{ \frac{1 - z^{-1}}{\log z} \right\}^{-1}$$

Proof Continued

So that we get,

$$\frac{dZ}{dt} = g\left(e^{\mathrm{ad}X} e^{t\mathrm{ad}\gamma}\right)(Y).$$

Observing that $Z(0) = X$ and integrating gives us

$$\log\left(e^X e^Y\right) = Z(1) = X + \int_0^1 g\left(e^{\mathrm{ad}X} e^{t\mathrm{ad}\gamma}\right)(Y)dt$$

which is the Baker-Campbell-Hausdorff formula.

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Corollary 2.1.1

(Series form of the BCH Formula) : For all $n \times n$ complex matrices X and Y with $\|X\|$ and $\|Y\|$ sufficiently small, we have

$$\log\left(e^X e^Y\right) = X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{1}{12}[Y, [X, Y]] + \cdots$$

Proof

We use the analytic series of $g(z)$

$$g(z) = 1 + \frac{1}{2}(z-1) - \frac{1}{6}(z-1)^2 + \frac{1}{12}(z-1)^3 - \cdots .$$

First of all,

$$\begin{aligned} & e^{\mathsf{ad}_X} e^{t\,\mathrm{ad}_Y} - I \\ &= \left(I + \mathsf{ad}_X + \frac{(\mathsf{ad}_X)^2}{2} + \cdots \right) \left(I + t\,\mathsf{ad}_Y + \frac{t^2\,(\mathsf{ad}_Y)^2}{2} + \cdots \right) - I \\ &= \mathsf{ad}_X + t\,\mathsf{ad}_Y + t\,\mathsf{ad}_X\,\mathsf{ad}_Y + \frac{(\mathsf{ad}_X)^2}{2} + \frac{t^2\,(\mathsf{ad}_Y)^2}{2} + \cdots . \end{aligned}$$

Computing up to degree 2 in ad_X and ad_Y gives

Proof Continued

$$g\left(e^{\mathrm{ad}_X} e^{t\,\mathrm{ad}_Y}\right) = I + \frac{1}{2}\left[\mathrm{ad}_X + t\,\mathrm{ad}_Y + t\,\mathrm{ad}_X\,\mathrm{ad}_Y + \frac{(\mathrm{ad}_X)^2}{2} + \frac{t^2(\mathrm{ad}_Y)^2}{2}\right]$$
$$- \frac{1}{6}\left[(\mathrm{ad}_X)^2 + t^2(\mathrm{ad}_Y)^2 + t\,\mathrm{ad}_X\,\mathrm{ad}_Y + t\,\mathrm{ad}_Y\,\mathrm{ad}_X\right] + \text{ higher-order terms.}$$

$$\begin{aligned}
&\log\left(e^X e^Y\right) \\
&= X + \int_0^1 \left[Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] - \frac{t}{6}[Y, [X, Y]]\right] dt + \cdots \\
&= X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] \\
&\quad - \frac{1}{12}[Y, [X, Y]] + \cdots
\end{aligned}$$

for first few terms.

Definition 3.1

If H is any subgroup of $GL(n;\mathbb{C})$. Then,

$$\mathfrak{h} = \{X \in \mathfrak{h} : e^{tX} \in H, \quad \forall t \in \mathbb{R}\}$$

is **lie algebra** of H.

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Definition 3.2

If G is a matrix Lie group with Lie algebra $\mathfrak{g}$, then $H \subset G$ is a **connected Lie subgroup** of G if the following conditions are satisfied:

1. H is a subgroup of G.
2. The Lie algebra $\mathfrak{h}$ of H is a Lie subalgebra of $\mathfrak{g}$.
3. Every element of H are in the form $e^{X_1}e^{X_2}\cdots e^{X_m}$, with $X_1, \ldots, X_m \in \mathfrak{h}$.

Observe that any group H as in the definition is path connected, since each element of H can be connected to the identity I in H by a path

$$t \mapsto e^{(1-t)X_1} e^{(1-t)X_2} \cdots e^{(1-t)X_m}$$

Lemma 2

Choose a basis for $\mathfrak{h}$ and call an element of $\mathfrak{h}$ **rational** if its coefficients with respect to this basis are rational. Then for every $\delta > 0$ and every $A \in H$, there exist rational elements $R_1, \ldots, R_m$ in $\mathfrak{h}$ and X in $\mathfrak{h}$ with $\|X\| < \delta$, such that

$$A = e^{R_1} e^{R_2} \cdots e^{R_m} e^{X}.$$

Proof

Choose $\varepsilon > 0$ such that $\forall \quad X, Y \in \mathfrak{h}$ with $\|X\| < \varepsilon$ and $\|Y\| < \varepsilon$, the Baker-Campbell-Hausdorff formula holds for X and Y. Let $C(\cdot,\cdot)$ denote the right-hand side of the formula, so that

$$e^X e^Y = e^{C(X,Y)}$$

whenever $\|X\|, \|Y\| < \varepsilon$. Also $C(\cdot,\cdot)$ is continuous in X, Y. So for a given ε there exists some $\delta > 0$ such that for $\|X\|, \|Y\| < \delta$, implies $\|C(X,Y)\| < \varepsilon$. So, we can assume $\delta < \varepsilon$. Since $e^X = \left(e^{X/k}\right)^k$, we can write every element A of H in the form

$$A = e^{X_1} \cdots e^{X_N}$$

with $X_j \in \mathfrak{h}$ and $\|X_j\| < \delta$.

Proof Continued

We now apply induction on N. If $N = 0$, then $A = I = e^0$, so nothing to prove. Assume the lemma for A 's that can be expressed in the above form for some integer N, and consider A of the form

$$A = e^{X_1} \cdots e^{X_N} e^{X_{N+1}}$$

with $X_j \in \mathfrak{h}$ and $\|X_j\| < \delta$. Applying our induction hypothesis to $e^{X_1} \cdots e^{X_N}$, we obtain

$$\begin{aligned} A &= e^{R_1} \cdots e^{R_m} e^{X} e^{X_{N+1}} \\ &= e^{R_1} \cdots e^{R_m} e^{C(X, X_{N+1})} \end{aligned}$$

where the R_j 's are rational and $\|C(X, X_{N+1})\| < \varepsilon$. Since $\mathfrak{h}$ is a subalgebra of $\mathrm{gl}(n; \mathbb{C})$, $C(X, X_{N+1})$ is in $\mathfrak{h}$, but it may not have norm less than δ. So choose a rational element R_{m+1} of $\mathfrak{h}$ that is close to $C(X, X_{N+1})$ with $\|R_{m+1}\| < \varepsilon$ as rational points are dense in $\mathfrak{h}$.

Proof Continued

So we get,

$$A = e^{R_1} \cdots e^{R_m} e^{R_{m+1}} e^{-R_{m+1}} e^{C(X, X_{N+1})}$$
$$= e^{R_1} \cdots e^{R_m} e^{R_{m+1}} e^{X'}$$

where $X' = C\left(-R_{m+1}, C\left(X, X_{N+1}\right)\right)$
Then X' will be in $\mathfrak{h}$, and by choosing R_{m+1} sufficiently close to $C\left(X, X_{N+1}\right)$, we can make $\|X'\| < \delta$. Since $C(-Z, Z) = \log\left(e^{-Z} e^{Z}\right) = 0$ for all small Z, and if Z' is close to Z, then $C\left(-Z', Z\right)$ will be more smaller.

Discussion:Think of $\mathrm{gl}(n;\mathbb{C})$ as $\mathbb{R}^{2n^2}$ and decompose $\mathrm{gl}(n;\mathbb{C})$ as $\mathfrak{h} \oplus D$, where D is the orthogonal complement of $\mathfrak{h}$ with respect to the usual inner product on $\mathbb{R}^{2n^2}$ as show in in the proof of Theorem (1.3).Then by the Inverse Function Theorem, there exist neighborhoods U and V of the origin in $\mathfrak{h}$ and D, respectively, and a neighborhood W of I in $\mathrm{GL}(n;\mathbb{C})$ with the property that for each $A \in W$ we can write it uniquely as

$$A = e^X e^Y, \quad X \in U, Y \in V \tag{3.2}$$

in such a way that X and Y depend continuously on A.

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Lemma 3

Following the above discussion let V be a neighborhood of the origiin in D as in (3.2).If $E \subset V$ is defined as

$$E = \{Y \in V| \quad e^Y \in H\}$$

Then E is at most countable.

Proof

Fix δ such that for all X , Y with $\|X\|, \|Y\| < \delta$, the quantity $C(X, Y)$ is defined and contained in U. Our claim is that for each sequence $R_1, \ldots, R_m$ of rational elements in $\mathfrak{h}$, there is at most one $X \in \mathfrak{h}$ with $\|X\| < \delta$ such that the element

$$e^{R_1} e^{R_2} \cdots e^{R_m} e^X \qquad (3.3)$$

belongs to e^V. Since, if we have

$$\begin{aligned} e^{R_1} e^{R_2} \ldots e^{R_m} e^{X_1} &= e^{Y_1}, \\ e^{R_1} e^{R_2} \ldots e^{R_m} e^{X_2} &= e^{Y_2} \end{aligned}$$

with $Y_1, Y_2 \in V$, then

$$e^{-X_1} e^{X_2} = e^{-Y_1} e^{Y_2}.$$

Proof Continued

So

$$e^{-Y_1} = e^{-X_1} e^{X_2} e^{-Y_2} = e^{C(-X_1, X_2)} e^{-Y_2}$$

with $C(-X_1, X_2) \in U$. However, each element of $e^U e^V$ has a unique representation as $e^Y e^X$ with $X \in U$ and $Y \in V$. Thus, we must have $-Y_2 = -Y_1$ and so, $e^{X_1} = e^{X_2}$ and $X_1 = X_2$. By Lemma (3.1), every element of H can be expressed in the (3.1) lemma form with $\|X\| < \delta$. Now, there are only countably many rational elements in $\mathfrak{h}$ and thus only countably many expressions of the form $e^{R_1} \cdots e^{R_m}$, each of which produces at most one element of the (3.3) form that belongs to e^V. Thus, the set E in Lemma (3.3) is at most countable.

Theorem 3.1

Let G be a matrix Lie group with Lie algebra $\mathfrak{g}$ and let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Then there exists a unique connected Lie subgroup H of G with Lie algebra $\mathfrak{h}$.

Proof

It is enough to assume that $G = \mathrm{GL}(n; \mathbb{C})$. Indeed, if G is a matrix lie group of $\mathrm{GL}(n; \mathbb{C})$ and $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$ then applying the theorem for $\mathrm{GL}(n; \mathbb{C})$ there exists H a connected Lie subgroup of $\mathrm{GL}(n; \mathbb{C})$ as $\mathfrak{h}$ is a lie subalgebra in $\mathfrak{gl}(n; \mathbb{C})$. And for every A in H written as $A = e^{X_1} e^{X_2} \cdots e^{X_m}$ with $X_i \in \mathfrak{h} \subset \mathfrak{g}$ we get that $A \in G$. Hence H is a connected lie subgroup of G.
Now let

$$H = \left\{ e^{X_1} e^{X_2} \cdots e^{X_N} \mid X_1, \ldots, X_N \in \mathfrak{h} \right\}$$

which is a subgroup of G. All we need to prove that the Lie algebra of H is $\mathfrak{h}$.

Proof Continued

Let $\mathfrak{h}'$ be the Lie algebra of H. Then $\mathfrak{h} \subseteq \mathfrak{h}'$. For $Z \in \mathfrak{h}'$, we may write, for all sufficiently small t,

$$e^{tZ} = e^{X(t)}e^{Y(t)} \qquad (3.4)$$

where $X(t) \in U \subset \mathfrak{h}$ and $Y(t) \in V \subset D$ and $e^{tZ} \in W \subset H$ by using (3.2).Note that $X(t)$ and $Y(t)$ are continuous functions of t as by the local inverse function (which exists due to the inverse function theorem), $X(t), Y(t)$ depends on tZ continuously and hence on t since Z is fixed. Since $Z \in \mathfrak{h}$, we have $e^{tZ} \in H$ for all t. Also $e^{X(t)} \in H$, hence $e^{Y(t)} \in H$ for all sufficiently small t. If $Y(t)$ is not constant, then it would take on uncountably many values, implying E is uncountable, contradicting Lemma (3.3) . So, $Y(t)$ must be constant. Since $Y(0) = 0$ which comes from the uniqueness of the above equation (3.4), $Y(t) \equiv 0$. Thus, for small t, we get $e^{tZ} = e^{X(t)}$ implying $tZ = X(t) \in \mathfrak{h}$. So, $Z \in \mathfrak{h}$ implying $\mathfrak{h}' \subseteq \mathfrak{h}$.

Theorem 4.1

Levi Decomposition Theorem: Let $\mathfrak{g}$ be a lie algebra over a field F and S be its radical.Then $\mathfrak{g}$ admits a Levi subalgebra that means, there exists a lie subalgebra H in $\mathfrak{g}$ s.t

$$\mathfrak{g} = S \oplus H$$

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Proposition 4.1

Let $L = S \oplus H$ be a finite-dimensional Lie algebra where S is a solvable ideal and H is a subalgebra of L. Suppose $\sigma : S \to \mathfrak{gl}(V)$ be a finite-dimensional nilrepresentation. Then there exists a finite-dimensional representation $\rho : L \to \mathfrak{gl}(V')$ such that $\ker(\rho) \cap S \subseteq \ker(\sigma)$. Also, if $\mathrm{Nil}(L) = \mathrm{Nil}(S)$ or $\mathrm{Nil}(L) = L$, then ρ can be chosen to be a nilrepresentation.

Theorem 4.2

Ado's Theorem: Let L be a finite-dimensional Lie algebra over a field of characteristic 0 . Then L has a faithful finite-dimensional representation.

Proof

Let $Z(L)$ denote the centre of L. Let $z_1, z_2, \ldots, z_n$ be a basis for $Z(L)$. We claim that the map $\rho_0 : Z(L) \to \mathfrak{gl}\left(F^{2n}\right)$ defined by

$$\begin{aligned}\rho_0\left(c_1 z_1 + c_2 z_2 + \cdots + c_n z_n\right) : &\left(a_1, b_1, a_2, b_2 \ldots, a_n, b_n\right)\\ &\longmapsto \left(c_1 b_1, 0, c_2 b_2, 0, \ldots, c_n b_n, 0\right)\end{aligned}$$

is a faithful finite-dimensional nilrepresentation of $Z(L)$. Indeed if

$$\rho_0\left(c_1 z_1 + c_2 z_2 + \cdots + c_n z_n\right) = \rho_0\left(d_1 z_1 + d_2 z_2 + \cdots + d_n z_n\right),$$

Proof Continued

then

$$(c_1b_1, 0, c_2b_2, 0, \ldots, c_nb_n, 0) = (d_1b_1, 0, d_2b_2, 0, \ldots, d_nb_p, 0)$$

for every $b_1, b_2, \ldots, b_p \in F$. Setting $b_i = 1$ and $b_j = 0$ for all $i \neq j$, we get $c_i = d_i$. So ρ_0 is faithful. To check that ρ_0 is a representation, we need to prove $\rho_0([x, y]) = [\rho_0(x), \rho_0(y)]$ for every $x, y \in Z(L)$. Since $Z(L)$ is commutative, this is equivalent to showing $[\rho_0(x), \rho_0(y)] = 0$, i.e. $\rho_0(x) \circ \rho_0(y) = \rho_0(y) \circ \rho_0(x)$. Let $x = c_1z_1 + \cdots + c_nz_n$ and $y = d_1z_1 + \cdots + d_nz_n$. Then,

$$\rho_0(y) \circ \rho_0(x) : (a_1, b_1, a_2, b_2 \ldots, a_n, b_n) \xrightarrow{\rho_{\mathbf{0}}(x)}$$

$$(c_1b_1, 0, c_2b_2, 0, \ldots, c_nb_n, 0) \xrightarrow{\rho_{\mathbf{0}}(y)} (0, 0, 0, 0 \ldots, 0, 0)$$

Proof Continued

and similarly,

$$\rho_0(y) \circ \rho_0(x) : (a_1, b_1, a_2, b_2 \ldots, a_n, b_n) \xrightarrow{\rho_{\mathbf{0}}(x)} (c_1 b_1, 0, c_2 b_2, 0, \ldots, c_n b_n, 0) \xrightarrow{\rho_{\mathbf{0}}(y)} (0, 0, 0, 0 \ldots, 0, 0)$$

So $\rho_0(x) \circ \rho_0(y) = \rho_0(y) \circ \rho_0(x) = 0$ as required. In particular, $(\rho_0(x))^2 = 0$ for all $x \in Z(L)$, which shows that ρ_0 is a nilrepresentation. Because $\text{Rad}(L)$ is solvable, and $Z(L) \subseteq \text{Rad}(L)$ there exists a sequence of solvable subalgebras(infact ideals) in L

$$Z(L) = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_k = \text{Nil}(L) \subseteq L_{k+1} \subseteq \cdots \subseteq L_m = \text{Rad}(L) \subseteq L_{m+1} = L$$

Proof Continued

such that L_i is an ideal in L_{i+1} for $0 \leq i \leq m$, and $\dim(L_{i+1}) = \dim(L_i) + 1$ for $0 \leq i \leq m-1$. Hence, $L_{i+1} = L_i \oplus Fv_i$ where $v_i \in L_{i+1} \backslash L_i$ for each $0 \leq i \leq m-1$. And for the case $i = m$, we use Levi's Theorem to get $L = \text{Rad}(L) \oplus H$ for some semisimple subalgebra H of L.

Proof Continued

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Starting with the nilrepresenation ρ_0 of $Z(L) = L_0$, we apply the last Proposition inductively to get a representation ρ_1 of L_1 such that $\ker(\rho_1) \cap L_0 \subseteq \ker(L_0)$. For the induction to work,we need each ρ_i is nilrepresentation which is true because $\text{Nil}(L_i) = L_i$ for $1 \leq i \leq k$ (as $L_i \subseteq \text{Nil}(L)$ for $0 \leq i \leq k$) and $\text{Nil}(L_i) = \text{Nil}(L)$ for $i \geqslant k$(as $\text{Nil}(L)$ is the maximal nilpotent ideal of L). Note that from $Z(L) = Z(L) \cap L_i$ for all $0 \leq i \leq m+1$ we get

$$Z(L) \cap \ker(\rho_i) = Z(L) \cap L_{i-1} \cap \ker(\rho_i) \subseteq Z(L) \cap \ker(\rho_{i-1})$$

Proof Continued

for all $1 \leq i \leq m+1$. Since $Z(L) \cap \ker(\rho_0) = 0$ (because ρ is faithful), it follows that $Z(L) \cap \ker(\rho_i) = 0$ for all $1 \leq i \leq m+1$. Hence, $Z(L) \cap \ker(\rho_{m+1}) = 0$. We have the representation $\rho_{m+1} : L_{m+1} = L \to \mathfrak{gl}(V)$ for some vector space with $\dim(V) < \infty$. Define $\rho = \rho_{m+1} \oplus \mathrm{ad} : L \to \mathfrak{gl}(V \oplus L)$ by $\rho(x) = (\rho_{m+1}, \mathrm{ad}(x))$. Then ρ is a finite-dimensional representation of L satisfying

$$\ker(\rho) = \ker(\rho_{m+1}) \cap \ker(\mathrm{ad}) = \ker(\rho_{m+1}) \cap Z(L) = 0$$

so ρ is faithful. Hence proved.

Lie's Third Theorem

Theorem 4.3

If $\mathfrak{g}$ is any finite dimensional real Lie Algebra, there exists a connected Lie subgroup G of $GL(n;\mathbb{C})$ whose lie algebra is isomorphic to $\mathfrak{g}$.

Proof

By Ado's theorem, we identify $\mathfrak{g}$ with a real lie subalgebra of $\mathfrak{gl}(n;\mathbb{C})$.Hence, by theorem (3.1), there is a connected lie subgroup of $GL(n;\mathbb{C})$ with lie algebra $\mathfrak{g}$.

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Thank You